Fiscal Year:	FY 2013	**Task Last Updated:**	FY 12/11/2013
PI Name:	Bruinsma, Douwe		
Project Title:	Computer-Controlled Force Generator		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Wheat Ridge	**State:**	CO
Zip Code:	80212-1916	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	01/02/2013	**End Date:**	01/01/2015
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13CC02C		
Performance Goal No.:			
Performance Goal Text:			
	TDA Research, Inc. is developing a compact, low power, Next-Generation Exercise Device (NGRED) that can generate any force between 5 and 600 lbf. We use a closed loop control system and a servo motor to smoothly and accurately simulate the gravitational and inertial loads of lifting a weight on earth. However, because the system uses a computer-controlled motor, the load can be varied independently during the concentric and eccentric phase of the exercise. Thus, the system can easily provide an eccentric overload during the return stroke, greatly increasing the physiological benefit of a workout. The NGRED has a user-friendly interface where the exercise is selected from a drop-down menu, along with the desired weight and overload. The NGRED will automatically adjust to the user's stored range of motion (ROM) for the selected exercise and apply the set load only during the ROM. The NGRED automatically applies the eccentric overload at the top of the ROM and advances to the next rep at the bottom. The time required to change between users, exercises, and weights is less than 10 seconds. This makes much better use of the astronaut's time; with current mechanically adjustable		

Task Description:	exercise machines up to two thirds of the time is spent adjusting the machine. The software includes data logging and communication abilities to meet NASA requirements as well as redundant hardware and software fail-safe mechanisms. The NGRED includes an efficient energy recovery system which stores the energy generated by the user during the concentric phase of the motion (pull stroke) and applies that energy to provide resistance during the eccentric phase (return stroke). The average power consumption of the NGRED will be less than 50 W during an exercise session. The expected weight of the NGRED at the end of Phase II will be 20 kg, and the total volume is expected to be 55 L, including all electronics and controls. A flight-like NGRED will be delivered to NASA at the end of Phase II. POTENTIAL NASA COMMERCIAL APPLICATIONS: Extended stays in reduced gravity environments lead to a decrease in bone density and muscular deterioration if proper countermeasures are not taken. Studies have shown that crewmembers of the International Space Station can lose up to 2% per month in bone mass and 32% in muscle strength during a 6 month stay. To counteract this phenomenon, several pieces of exercise equipment have been developed. These pieces range from elastic bands to more complex pieces of equipment such as the interim Resistive Exercise Device (iRED) and the Advanced Resistive Exercise Device (ARED). The iRED does not generate sufficient force for effective resistive exercise, whereas the ARED has been proven to be sufficient for maintaining muscle mass and bone density. The ARED, however, is large and heavy, making it unsuitable for incorporation into a small spacecraft for long duration space travel. The NGRED is able to generate sufficient resistive force for effective resistive exercise (up to 600 lbf) and is able to provide eccentric overload to further increase the efficiency of a resistive exercise session. Furthermore, the NGRED is lightweight, low power, compact, and user friendly, making it perfectly suitable for inclusion in a small spacecraft for long-duration missions. Average power consumption of the NGRED will be less than 50 W during operation, the volume will be less than 55 L and the weight will be less than 20 kg.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The NGRED has markets in academic research, physical therapy, athletic departments, and fitness industry. The NGRED can be used in home-gyms to replace the heavy weight-stacks and add an electronic interface to monitor progress or to share results with a remote coach or friends online. There is much research being performed to study the physiological benefits of different load profiles during strength training, with the majority of this work being focused on the effects of eccentric overload. The NGRED is perfectly suited for this because the amount of eccentric overload can be precisely controlled and set by entering the desired value on the user-interface. The NGRED also allows the study of custom load profiles throughout an exercise motion. For this application the NGRED has unique capabilities in that can match a user's range of motion (ROM) in seconds and then apply a custom load profile based on the ROM while logging position and force data at 16 kHz. Lastly, the features of the NGRED make it ideal for physical therapy centers. With the NGRED, strength training can be designed to precisely meet the needs of the patient by providing resistive loads only where desired during the range of motion. To market our device, we will use a proven commercialization pathway, building credibility through sports medicine research and incorporation into physical therapy facilities, leading to large sales in athletic departments, fitness centers, and ultimately in home markets.
Task Progress:	New project for FY2013. Reporting not required for this SBIR Phase 2 project.
Bibliography Type:	Description: (Last Updated:)